2. Assignment on Numpy

1. Create a NumPy array containing integers from 0 to 9.
2. Create a NumPy array from a given Python list [10, 20, 30, 40, 50].
3. Create a 2D array (3x3) with values ranging from 0 to 8.
4. Given a 2D array, find its shape and size. Create a 5x4 array with random integers between 1 and 100, and print the shape and size.
5. Create a 3x3 array of integers and access:
    a. The second element of the first row.
    b. The third column of the 2nd row.
    c. A subarray with the first two rows and last two columns.
6. Create a 1D array of 12 elements. Reshape it into a 3x4 2D array.
7. Reshape a 3x3 matrix into a 1D array.
8. Create two arrays of shape (2, 3): arr1 and arr2. Perform element-wise addition, subtraction, multiplication, and division on them.
9. Use NumPy's built-in functions to compute the sum, mean, and standard deviation of a given array.
10. Create a 2x2 matrix A and a 2x2 matrix B. Perform matrix multiplication using np.dot() or the @ operator.
11. Find the transpose of matrix A and matrix B.
12. Create a 3x3 array and a 1D array with 3 elements. Add the 1D array to the 2D array. Explain how broadcasting works in this operation.
13. Create a 4x4 array of random integers. Find the maximum, minimum, and mean of the entire array.
14. Find the row-wise and column-wise sums.
15. Generate an array of 10 random integers between 1 and 50.
16. Generate a 5x5 matrix of random floats between 0 and 1.

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